

NYC Health + Hospitals

OBGYN — AI Scheduling Platform

User Guide & Demonstration Walkthrough

FOR DEMONSTRATION ONLY — NOT FOR CLINICAL USE — SYNTHETIC DATA

This guide walks an operator through the demo's functionality — role-based panes, no-show risk, PTO impact & coverage conflicts, the human-in-the-loop approval gate, the natural-language scheduling assistant, and the Epic/MCP data layer. Each section gives a step-by-step **procedure** and the **expected outcome**.

All data is synthetic. No PHI is used. Names are fictional; phone numbers use the reserved 555-01xx range; MRNs are SYN-xxxx. Every AI recommendation is human-approved and audited.

Application UI	https://nychhc-frontend-iis-ai-ui.apps.ocp419.crucible.iisl.com
Backend API	https://nychhc-copilot-iis-ai-ai.apps.ocp419.crucible.iisl.com
Dashboards	Grafana — https://grafana-iis-ai-ui.apps.ocp419.crucible.iisl.com (admin / Demo1234#)
Platform	OpenShift AI (ocp419.crucible.iisl.com) · CPU-only · synthetic data

AI solution by IIS · iistech.com

1 · Getting started — access & roles

The demo tailors itself to four roles. Pick a role in the left sidebar; the available tabs and the assistant's permissions change to match.

Procedure

1. Open the Application UI in a browser. The cluster uses a self-signed certificate — accept the browser warning and continue.
2. In the left sidebar under **Role & context**, choose a role: **Scheduler** (persona Selamawit), **Approver**, **Provider**, or **Leadership**.
3. Note the tab bar updates for the role, and the amber **FOR DEMONSTRATION ONLY** banner is shown on every screen.

✓ EXPECTED OUTCOME

Scheduler → Dashboard, Schedule, No-Show Risk, Approvals, Assistant.

Approver → Dashboard, PTO, Approvals, Assistant.

Provider → Schedule, PTO, Assistant (read-only for writes).

Leadership → Reporting, Dashboard, Assistant.

2 · No-show risk (UC1)

Each appointment gets a no-show probability and a red / amber / green tier with the human-readable factors behind it, so outreach can be prioritised before a slot is wasted.

Procedure

1. As **Scheduler**, open the **No-Show Risk** tab.
2. Use the All / Red / Amber / Green filter, or search by patient, MRN, or provider.
3. Read the **Top factors** column for each flagged appointment and the recommended action.

✓ EXPECTED OUTCOME

A risk-scored list for today: RED > 65%, AMBER 35–65%, GREEN < 35% (thresholds are configurable without code).

Example: **Mei Chen** 75% RED — “3 prior no-shows, New OB, no text on file” → Call patient.

If the No-Show model endpoint is unreachable, a **degraded-mode** banner appears and rules-based scores are shown — never fabricated model output.

3 · Schedule drill-down — book / modify / cancel

The scheduling drawer walks specialty → doctor → calendar → confirm. The same actions the UI performs are also available to the assistant (one source of truth).

Procedure

1. As **Scheduler**, open the **Schedule** tab and click **New appointment**.
2. Choose a specialty (e.g. Obstetrics), then a provider (e.g. Dr. Amara Okonkwo), then an open slot on the calendar, then confirm.
3. To change one, use **Modify** or **Cancel** and search by patient name or MRN.

✓ EXPECTED OUTCOME

Open / Booked / Blocked slots are computed from the live schedule; double-booking is rejected.

Cancelling frees the slot and suggests higher-risk patients to re-offer it to.

4 · PTO impact & coverage conflict (UC4)

Before a leave request is approved, the system shows the downstream appointment impact and flags coverage conflicts — including overlap with another provider's leave.

Procedure

1. As **Approver**, open the **PTO** tab and select the pending request for **Dr. Amara Okonkwo** (Jun 16–20).
2. Review the impacted appointments and the auto-resolvable vs. manual split.
3. Read the coverage-conflict banner and the recommended mitigation.

✓ EXPECTED OUTCOME

4 appointments impacted; 3 auto-resolvable (reassign to a peer obstetric provider).

Coverage conflict: Inpatient OB needs 2 on service but falls short on 3 days — overlaps with Dr. Rachel Stein's leave. Mitigation: stagger leave, pull a peer onto service, or deny one request.

Approving a request that leaves an uncovered window is **blocked** until explicitly overridden.

5 · Human-in-the-loop approvals & audit (UC6)

Nothing the AI proposes executes on its own. Every recommendation is staged for a human decision, and every decision is recorded with the user and a timestamp.

Procedure

1. Trigger a recommendation — e.g. on the **PTO** tab click **Apply all auto**, or ask the assistant to apply reassignments (see section 6).
2. Open the **Approvals** tab (Scheduler or Approver).
3. In **Pending approvals**, click **Approve** or **Reject**.
4. Scroll to the **Audit trail** to see the recorded decision.

✓ EXPECTED OUTCOME

On Approve, the action executes and a row is written: action, decision, outcome, the deciding **user**, and a **timestamp** (e.g. *pto_reassign · approved · executed · chat:Scheduler*).

On Reject, nothing executes and the rejection is logged. A Provider cannot approve writes.

6 · Scheduling Assistant — natural language (UC5)

Ask plain-English questions; a deterministic router answers the headline asks from real data (no LLM needed), and the assistant now remembers the conversation so follow-ups work.

Procedure

1. Open the **Assistant** tab. Use a suggested chip or type a question.
2. Try each of the three headline asks below, then the follow-up.
3. Use **Clear** to start a fresh conversation (also resets server-side context).

Try these:

Which OB providers have openings?
What's the no-show rate by provider?
Put Dr. Okonkwo on PTO 6/16-6/20 and show the impact
apply all auto (follow-up — uses the previous answer)

✓ EXPECTED OUTCOME

Openings → Obstetrics providers with soonest open slots (Stein 06-09 08:00, Okonkwo 08:20, Rahman 09:00).

No-show rate → average risk per provider with the high-risk (red) count.

PTO impact → the same impact + coverage conflict as the PTO tab.

“apply all auto” → remembers the prior turn and applies the reassignments through the approval gate: “Done — approved and applied 3 reassignment(s)... recorded in the audit trail.”

Guardrails

1. Ask “*who can cover Dr. Okonkwo?*” (no date) → the assistant asks a clarifying question rather than guessing.
2. Ask “*show me the nursing schedule*” → the assistant declines (out of scope) and states the boundary.

✓ EXPECTED OUTCOME

Answers are grounded in real data; ambiguous asks are clarified; out-of-scope and clinical/PHI requests are declined.

7 · Epic data via the MCP adapter (UC8)

The AI never touches Epic directly — it calls FHIR-shaped tools through a single adapter. In the demo the adapter is backed by synthetic, FHIR-mirrored data; swapping in a real Epic FHIR endpoint needs no caller changes.

Procedure

1. List the available tools (any HTTP client; -k for the self-signed cert):

```
curl -k <BACKEND>/api/mcp/tools
```

2. Invoke one tool and inspect the FHIR-shaped result:

```
curl -k -X POST <BACKEND>/api/mcp/call \
-H 'content-type: application/json' \
-d '{"tool": "get_provider_schedule", "args": {"provider_id": "p1", "date": "2026-06-09"}}'
```

✓ EXPECTED OUTCOME

/api/mcp/tools lists `get_patient_appointments`, `get_provider_schedule`, `check_slot_availability`, `get_risk_scores`, `check_pto_impact`.

/api/mcp/call returns FHIR R4 Appointment/Slot JSON; an unknown tool or unreachable source returns a typed error with *degraded: true* (no fabricated data).

8 · Leadership reporting & dashboards

The business case is department-level — throughput, coverage reliability, and capacity utilisation — for the Chair / CCO.

Procedure

1. Switch to the **Leadership** role and open the **Reporting** tab.
2. For operational charts, open **Grafana** and the **NYCHHC — Workforce & Patient-Flow** dashboard (sign in admin / Demo1234#).

✓ EXPECTED OUTCOME

Reporting shows coverage reliability, at-risk appointments, open coverage conflicts, and capacity utilisation tiles.

Grafana shows the no-show risk mix, no-show rate by provider, the PTO queue, and the roster.

9 · Reset between demos & troubleshooting

Procedure

1. In the Assistant, click **Clear** to reset the conversation (and server-side memory).
2. To rebuild operational data from scratch, an operator runs `./destroy.sh` then `./deploy.sh` (re-seeds the synthetic OBGYN data).

Notes

1. Models: if the KServe no-show/forecast endpoints are down, the demo automatically falls back to rules and shows the degraded banner — it never hard-fails.
2. Open-ended chat (beyond the headline asks) uses Claude via Portkey; the deterministic router still answers the headline asks even with no LLM key.
3. Replace <BACKEND> in the commands above with the Backend API URL on the cover.

FOR DEMONSTRATION ONLY — NOT FOR CLINICAL USE — SYNTHETIC DATA

AI solution by IIS · iistech.com — demonstration build, synthetic data only.